

Influence of Dissipative Forces and the Design of Mecanum-Wheels on the Omnidirectional Platform Dynamics

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Abstract—The object of research is an omnidirectional mobile platform equipped with four mecanum wheels. The movement of the system on a horizontal plane is considered. The aim of the research is to study the influence of the dissipative forces on the dynamics of the omnidirectional platform, taking into account the design of mecanum wheels: the shape of the rollers and their finite number. The equations of motion of the omnidirectional mobile platform are derived taking into account the real design of the mecanum wheels, their slippage, and the forces of multicomponent contact friction. A comparative analysis of the results of numerical modeling for various models of contact friction and different values of the parameters of dissipative forces is performed.

Index Terms—mecanum-wheel, mecanum-platform, Ilon wheel, omni-wheel, omni-platform, multicomponent friction

I. INTRODUCTION

The object of research is a mobile platform for omnidirectional movement, equipped with mecanum wheels (Swedish wheels or Ilon wheels [1]). On the periphery of each of them several rollers are mounted, the axes of which skew with the axis of the wheel at the angle of 45° . The design of the mecanum wheels allows the vehicle to make omnidirectional movement (i.e., in any direction with an arbitrary orientation), for example, to make translation motion along an arbitrary trajectory or to make a turn in place.

Due to their mobility and ease of parking, omnidirectional platforms are used to work in the cramped conditions of storage, production and similar premises; creating vehicles for disabled persons and for other purposes [2].

A mathematically rigorous description of the geometry of the mecanum wheels and their kinematics is given in [3]. The influence of the real design of the wheels on motion of omnidirectional platforms was studied in [4]–[9].

II. DESCRIPTION OF THE SYSTEM

In order to describe the kinematics of a mobile robot, we introduce a fixed coordinate frame xyz with a horizontal plane xy , and a moving coordinate frame $OXYZ$ with origin at

the geometric center of the platform O . The axis OX is a longitudinal symmetry axis of the platform, and the OY axis is a transverse axis. The coordinate axes z and OZ are vertical and aligned with each other. The kinematic scheme of the mecanum wheeled platform is shown at Fig. 1.

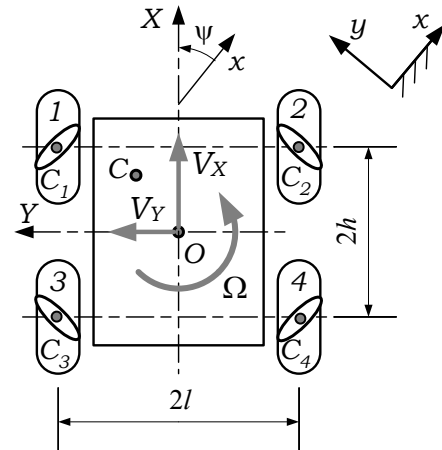


Fig. 1. Kinematic scheme of the mobile platform.

The position of the platform is defined by the Cartesian coordinates x_O and y_O of its geometric center O in fixed frame xyz and by the angle $\psi = \angle(x, OX)$.

The platform's center of mass C is displaced relative to the point O . Its Cartesian coordinates in moving plane OXY are denoted by a_X and a_Y respectively (see Fig. 1).

The *youBot* mobile robot is equipped by four pairwise coaxial mecanum wheels with six peripheral rollers. In the kinematic scheme of the platform, the rollers in contact with the floor are shown as ovals on wheels 1, 2, 3, 4.

The angles of the i -th wheel ($i = \overline{1, 4}$) rotation relative to the robot base are denoted by φ_i , and the angle of rotation of the contacting roller — by γ_i (see Fig. 2).

The angle between the XZ plane and the unit vector e_i for

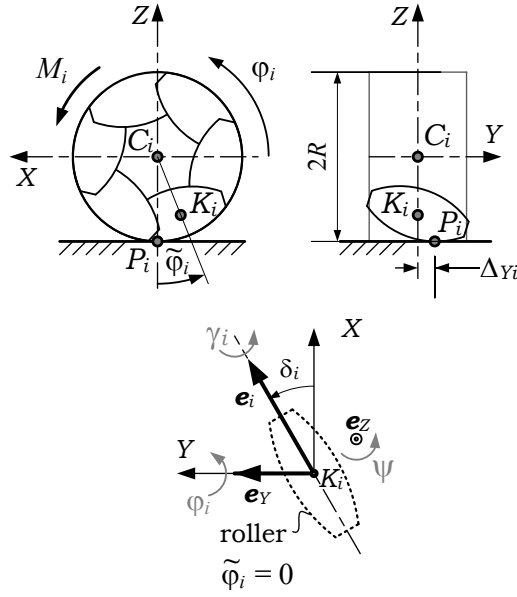


Fig. 2. Kinematic scheme of the i -th wheel

the i -th wheel contacting roller axis is equal to δ_i :

$$\delta_i = \angle(XZ, e_i) = \angle(X, e_i|_{\tilde{\varphi}=0}), \quad |\delta_i| = 45^\circ.$$

The Cartesian coordinates of the i -th wheel center C_i in the OXY plane are denoted by ρ_{Xi} and ρ_{Yi} .

Accordinging to the kinematic scheme (Fig. 1)

$$\begin{aligned} -\delta_1 &= \delta_2 = \delta_3 = -\delta_4 = 45^\circ, \\ \rho_{X1} &= \rho_{X2} = -\rho_{X3} = -\rho_{X4} = h, \\ \rho_{Y1} &= -\rho_{Y2} = \rho_{Y3} = -\rho_{Y4} = l. \end{aligned}$$

All rollers of each wheel contact internally a cylindrical surface of radius R , coaxial to the wheel. Denote by ΔY_i the displacement of the P_i contact point relative the wheel center C_i in direction of the Y axis (see Fig. 2) [3], [10]:

$$\Delta Y_i = R_d \tan \tilde{\varphi}_i \tan \delta_i, \quad (1)$$

$$\tilde{\varphi}_i = \left((\varphi_i + 30^\circ) \bmod 60^\circ \right) - 30^\circ. \quad (2)$$

Here $\tilde{\varphi}_i$ — the angle of rotation of the i -th wheel contacting roller about the Y axis (Fig. 2); $R_d = |C_i K_i|$ is distance between points C_i and K_i . As each wheel is equipped by six rollers, $-30^\circ \leq \tilde{\varphi}_i < 30^\circ$. If the contact roller switching occurs, $\tilde{\varphi}_i$ jumps from 30° to -30° or from -30° to 30° .

Further, if necessary, the brief notation $\sin = s, \cos = c$ is used.

III. EQUATIONS OF DYNAMICS OF THE SYSTEM AND CONTACT FRICTION MODELS

Earlier in the work of the authors [11], using the Appel's formalism, the equations of the platform dynamics were con-

structed taking into account the design of mecanum wheels in the form:

$$\begin{aligned} m_s(\dot{V}_X - \Omega V_Y) - m(a_Y \dot{\Omega} + a_X \Omega^2) &= \sum F_{Xi}, \\ m_s(\dot{V}_Y + \Omega V_X) + m(a_X \dot{\Omega} - a_Y \Omega^2) &= \sum F_{Yi}, \\ I_s \dot{\Omega} + m a_X (\dot{V}_Y + \Omega V_X) - m a_Y (\dot{V}_X - \Omega V_Y) &= \\ = \sum [F_{Yi} \rho_{Xi} - F_{Xi} (\rho_{Yi} + \Delta Y_i) + M_{Zi}], \\ J_1 \ddot{\varphi}_i &= M_i - \mu_1 \dot{\varphi}_i - F_{Xi} R, \\ J_r \ddot{\gamma}_i &= -\mu_2 \dot{\gamma}_i - F_{Xi} [s \delta_i (R - R_d c \tilde{\varphi}_i) - \Delta Y_i s \tilde{\varphi}_i c \delta_i] - \\ &\quad - F_{Yi} c \delta_i (R_d - R c \tilde{\varphi}_i) - M_{Zi} s \tilde{\varphi}_i c \delta_i, \quad i = \overline{1, 4}. \end{aligned} \quad (3)$$

where $m_s = m + 4m_1$ and I_s are the total mass of the robot and its total moment of inertia relative to the OZ axis; m is the platform mass; m_1 and J_1 are the wheel mass and inertia moment relative to the $C_i Y$ axis; J_r is the i -th roller inertia moment relative to the e_i axis; F_{Xi}, F_{Yi} are the projections of the friction force acting on the contacting roller of the i -th wheel reduced to the centers of the contact spots P_i ; M_{Zi} is the spin friction torque acting on the contacting roller of the i -th wheel; μ_1 and μ_2 are coefficients of the linear friction torques in base-wheel joints and wheel-roller ones; M_i are the actuator torques.

When constructing the equations of motion, the following were chosen as pseudo-velocities: the longitudinal $V_X = \dot{x}_O \cos \psi + \dot{y}_O \sin \psi$ and transverse $V_Y = \dot{y}_O \cos \psi - \dot{x}_O \sin \psi$ velocities of the center of the platform O , the angular velocity $\Omega = \dot{\psi}$ of the platform, the angular velocity $\dot{\varphi}_i$ of rotation of the i -th wheel and the angular velocity $\dot{\gamma}_i$ of rotation of the roller around its own axis

Consider as friction models:

- the Coulomb friction at the points of contact P_i :

$$\begin{aligned} F_{Xi} &= \frac{-f N_i V_{PiX}}{\sqrt{V_{PiX}^2 + V_{PiY}^2}}, \quad F_{Yi} = \frac{-f N_i V_{PiY}}{\sqrt{V_{PiX}^2 + V_{PiY}^2}}, \quad (4) \\ M_{Zi} &= 0, \end{aligned}$$

where f is the coefficient of sliding friction, N_i is the normal reaction force;

- multicomponent contact friction model [12]:

$$\begin{aligned} F_{Xi} &= -f N_i \frac{V_{PiX}}{\frac{8}{3\pi} \chi |\omega_{Zi}| + \sqrt{V_{PiX}^2 + V_{PiY}^2}}, \\ F_{Yi} &= -f N_i \frac{V_{PiY}}{\frac{8}{3\pi} \chi |\omega_{Zi}| + \sqrt{V_{PiX}^2 + V_{PiY}^2}}, \quad (5) \\ M_{Zi} &= -\frac{f N_i \chi^2 \omega_{Zi}}{\frac{16}{3\pi} \chi |\omega_{Zi}| + 5 \sqrt{V_{PiX}^2 + V_{PiY}^2}}, \end{aligned}$$

where χ is the contact spot radius; $\omega_{Xi} = \dot{\gamma}_i c \tilde{\varphi}_i c \delta_i$, $\omega_{Yi} = \dot{\varphi}_i - \dot{\gamma}_i s \delta_i$, $\omega_{Zi} = \Omega - \dot{\gamma}_i s \tilde{\varphi}_i c \delta_i$ are the projections of the vector of the angular velocity of the roller of the i -th wheel.

Note that in the multicomponent contact friction model, the contact spot of the rollers with the floor is represented by circles in which the distribution of contact stresses obeys

Hertz's law [13]. This takes into account slippage and spin of the contacting rollers around the axis Z .

The values of normal reactions present in friction models can be obtained in several ways, one of which is to solve a statically indefinable problem [11]. To close the models of friction forces, we present the expressions for the velocities of points P_i [9]–[11]:

$$\begin{aligned} V_{P_i X} &= V_X - \rho_{Y_i} \Omega - \dot{\varphi}_i R + \dot{\gamma}_i s \delta_i (R_d c \tilde{\varphi}_i - R) - \\ &\quad - \Delta_{Y_i} \cdot (\Omega - \dot{\gamma}_i s \tilde{\varphi}_i c \delta_i), \\ V_{P_i Y} &= V_Y + \rho_{X_i} \Omega + \dot{\gamma}_i c \delta_i (R c \tilde{\varphi}_i - R_d). \end{aligned} \quad (6)$$

IV. THE PLATFORM MOTION SIMULATION

A. Considered models of the system and input data for simulation

We analyze the effect of the parameters of dissipative forces on the movement of the platform, taking into account the real design of the mecamum wheels and contact friction, comparing the results of the numerical solution of the equations of motion for different values of these parameters for the following models of the system:

- **Model N** is non-holonomic. Simplifications are used: at each moment of time, the contact points P_i lie on the central cross sections of the rollers ($\tilde{\varphi}_i \equiv 0$, $\Delta_{Y_i} \equiv 0$), there is no slippage ($V_{P_i X} \equiv 0$, $V_{P_i Y} \equiv 0$). This model is described in [10].
- **Models CF** and **MF** are given by the dynamics equations (3) with the following contact forces:
 - Coulomb friction in contact points P_i (4) (**model CF**)
 - Multicomponent friction (5) (**model MF**).

The initial conditions for variables and the control torques M_i , $i = \overline{1, 4}$, are chosen the same for all three models under consideration using the following algorithm:

- 1) We set the desired platform motion

$$V_X = V_X^*(t), V_Y = V_Y^*(t), \Omega = \Omega^*(t).$$

- 2) By substituting $V_X = V_X^*(t)$, $V_Y = V_Y^*(t)$, $\Omega = \Omega^*(t)$ in the **model N** equations, the dependencies $M_i = M_i^*(t)$ are determined.
- 3) The found torque change laws $M_i = M_i^*(t)$, $i = \overline{1, 4}$, as well as initial conditions $V_X(0) = V_X^*(0)$, $V_Y(0) = V_Y^*(0)$, $\Omega(0) = \Omega^*(0)$, $\dot{\varphi}(0) = \dot{\varphi}_i^*(0)$, $\dot{\gamma}_i(0) = \dot{\gamma}_0^*(0)$ are used for **models CF** and **MF**.

The used numerical values of system parameters are presented in [9], [14], [15]. However, for the parameters of dissipative forces, consider some cases.

- **Case 1:** values taking from [16] for coefficients of the linear friction torques in base-wheel joints and wheel-roller ones (i.e $\mu_1 = 0.11 \text{ Nm} \cdot \text{s}$, $\mu_2 = 5.78 \cdot 10^{-3} \text{ Nm} \cdot \text{s}$) and the values of the contact spot radius: $\chi = 2 \text{ mm}$, $\chi = 5 \text{ mm}$, and $\chi = 8 \text{ mm}$.
- **Case 2:** “small” values for coefficients of the linear friction torques ($\mu_1 = 0.011 \text{ Nm} \cdot \text{s}$, $\mu_2 = 5.78 \cdot 10^{-4} \text{ Nm} \cdot \text{s}$)

and the values of the contact spot radius: $\chi = 2 \text{ mm}$, $\chi = 5 \text{ mm}$, and $\chi = 8 \text{ mm}$.

We consider as the desired motion uniform translational movement of the platform on a circle:

$$\begin{aligned} V_X^*(t) &= -\varrho \omega \cos(\omega t - \psi_0), V_Y^*(t) = \varrho \omega \sin(\omega t - \psi_0), \\ \Omega^*(t) &= 0, \end{aligned}$$

where $\varrho = 2.5 \text{ m}$ is the circle radius; $\omega = 2\pi/T$ is the cyclic frequency, and $T = 30 \text{ s}$ is the period of motion; $\psi_0 = 90^\circ$ is a constant angle of rotation of the platform ($\psi(t) = \psi_0$).

B. Simulation results for case 1

The control torques on such a movement change with the frequency ω according to sinusoidal laws (see Fig. 3).

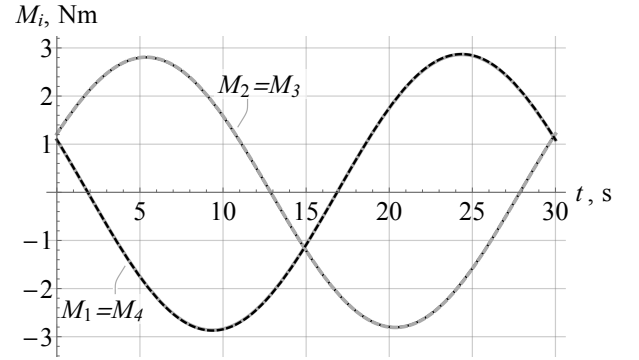


Fig. 3. Control torques for case 1

Fig. 4–6 show graphs of changes in the platform velocities ($0 \leq t \leq T$). Simulation results for the **model MF** with significant linear friction in the joints show, that the value of the contact spot radius has slightly effect on the movement of the mecamum-platform.

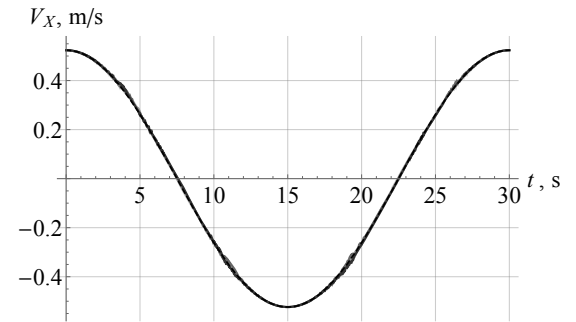


Fig. 4. Dependencies $V_X(t)$ for case 1 (**model N** — black dashed line; **model CF** — light gray solid line; **model MF**: $\chi = 2 \text{ mm}$ — gray solid line, $\chi = 5 \text{ mm}$ — dark gray solid line, $\chi = 8 \text{ mm}$ — black solid line)

The trajectories of the platform center O and the rotation angles ψ for three considered models of the system and different values of contact spot radius are shown in Fig. 7(traversing counterclockwise) and Fig. 8.

Using one wheel as an example, we present the dependencies of sliding friction (Fig. 9), sliding speed of the contact

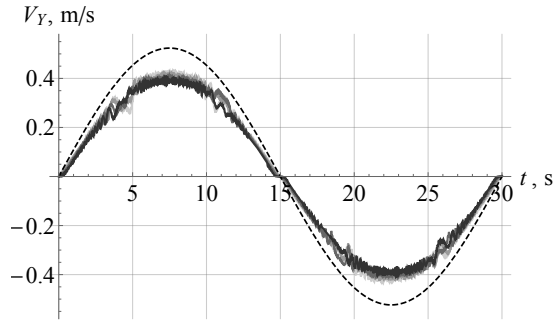


Fig. 5. Dependencies $V_Y(t)$ for **case 1** (**model N** — black dashed line; **model CF** — light gray solid line; **model MF**: $\chi = 2$ mm — gray solid line, $\chi = 5$ mm — dark gray solid line, $\chi = 8$ mm — black solid line)

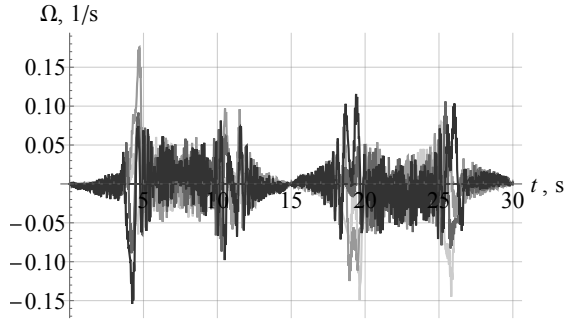


Fig. 6. Dependencies $\Omega(t)$ for **case 1** (**model N** — black dashed line; **model CF** — light gray solid line; **model MF**: $\chi = 2$ mm — gray solid line, $\chi = 5$ mm — dark gray solid line, $\chi = 8$ mm — black solid line)

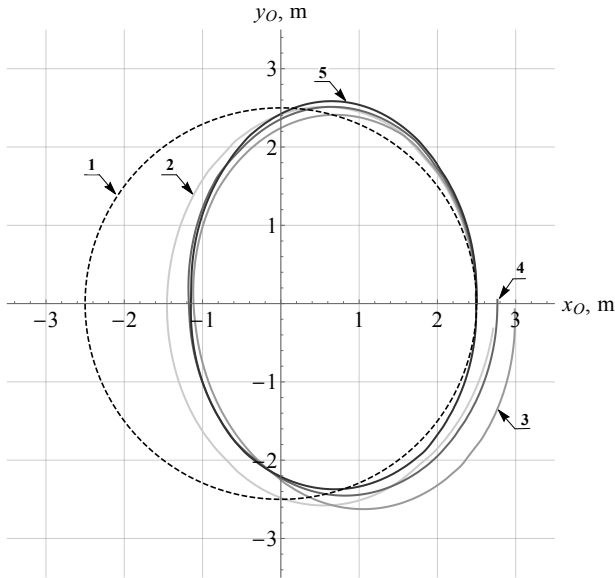


Fig. 7. Trajectories of the center of the platform O for **case 1** (1 — **model N**; 2 — **model CF**; 3 — **model MF**, $\chi = 2$ mm; 4 — **model MF**, $\chi = 5$ mm; 5 — **model MF**, $\chi = 8$ mm)

point (Fig. 10) and spin friction torque (Fig. 11). Note that the sliding velocities and the spin friction torque increase with an increase in the radius of the contact spot.

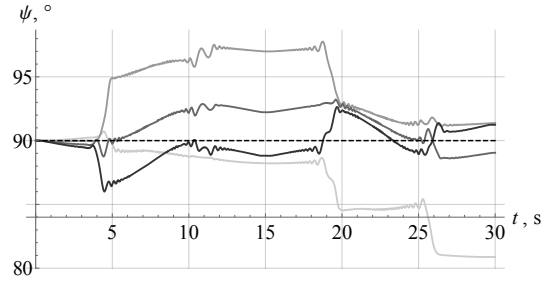


Fig. 8. Rotation angle of the platform $\psi(t)$ for **case 1** (**model N** — black dashed line; **model CF** — light gray solid line; **model MF**: $\chi = 2$ mm — gray solid line, $\chi = 5$ mm — dark gray solid line, $\chi = 8$ mm — black solid line)

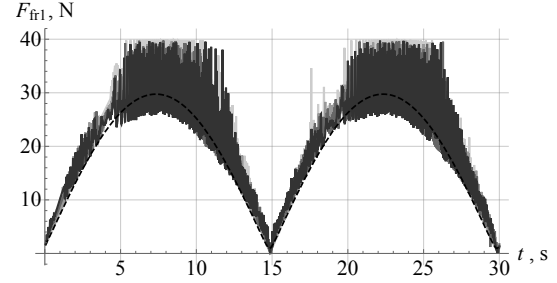


Fig. 9. Dependencies sliding friction force modulus $F_{fr1} = \sqrt{F_{X1}^2 + F_{Y1}^2}$ for **case 1** (**model N** — black dashed line; **model CF** — light gray solid line; **model MF**: $\chi = 2$ mm — gray solid line, $\chi = 5$ mm — dark gray solid line, $\chi = 8$ mm — black solid line)

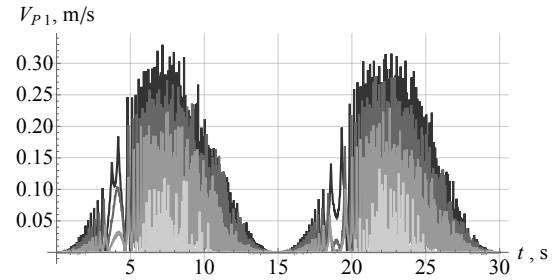


Fig. 10. Dependencies sliding speed modulus $V_{P1} = \sqrt{V_{P1X}^2 + V_{P1Y}^2}$ for **case 1** (**model CF** — light gray solid line; **model MF**: $\chi = 2$ mm — gray solid line, $\chi = 5$ mm — dark gray solid line, $\chi = 8$ mm — black solid line)

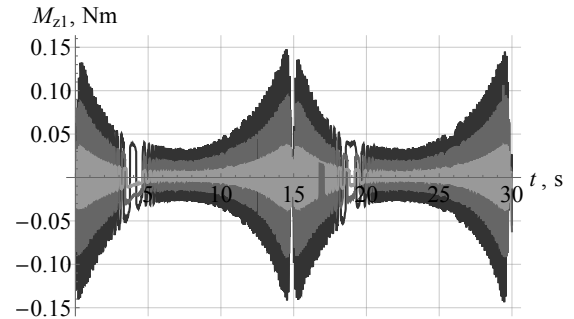


Fig. 11. Dependencies spin friction torque M_{Z1} for **case 1** according to the simulation results for **model MF** ($\chi = 2$ mm — gray solid line, $\chi = 5$ mm — dark gray solid line, $\chi = 8$ mm — black solid line)

C. Simulation results for case 2

The control torques on a movement change with the frequency ω according to sinusoidal laws (see Fig. 12).

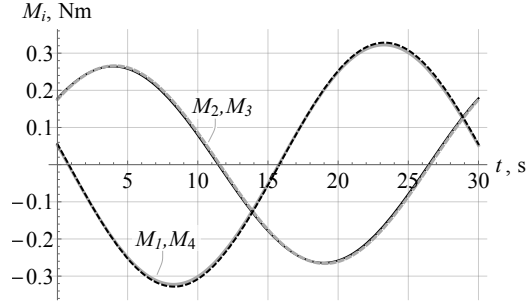


Fig. 12. Control torques for case 2

Fig. 13–16 show graphs of changes in the platform velocities and of the rotation speed of the first wheel ($0 \leq t \leq T$). Simulation results for the **model MF** with “small” linear friction in the joints show, that the value of the contact spot radius has significant effect on the movement of the mecatron-platform. Lateral movement efficiency decreases with increasing contact spot radius (Fig. 14, 15)

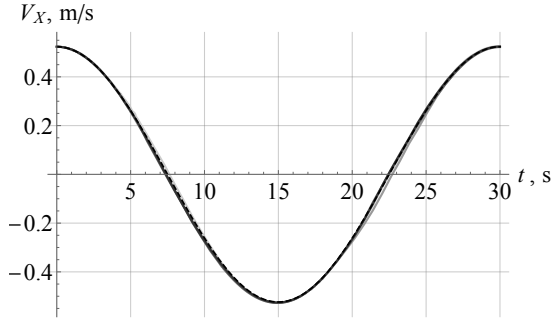


Fig. 13. Dependencies $V_X(t)$ for case 2 (**model N** — black dashed line; **model CF** — light gray solid line; **model MF**: $\chi = 2$ mm — gray solid line, $\chi = 5$ mm — dark gray solid line, $\chi = 8$ mm — black solid line)

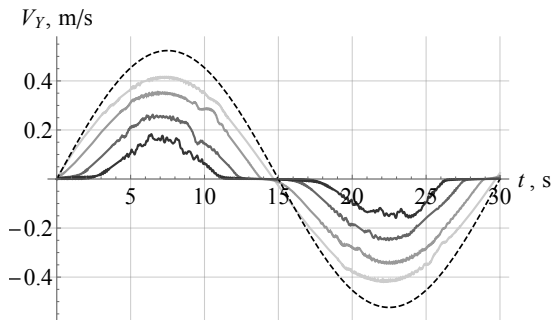


Fig. 14. Dependencies $V_Y(t)$ case 2 (**model N** — black dashed line; **model CF** — light gray solid line; **model MF**: $\chi = 2$ mm — gray solid line, $\chi = 5$ mm — dark gray solid line, $\chi = 8$ mm — black solid line)

The trajectories of the platform center O and the rotation angles ψ for three considered models of the system and

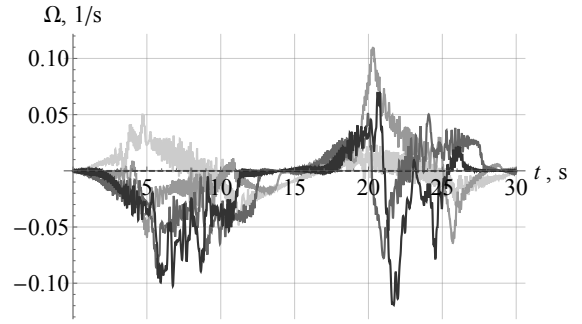


Fig. 15. Dependencies $\Omega(t)$ case 2 (**model N** — black dashed line; **model CF** — light gray solid line; **model MF**: $\chi = 2$ mm — gray solid line, $\chi = 5$ mm — dark gray solid line, $\chi = 8$ mm — black solid line)

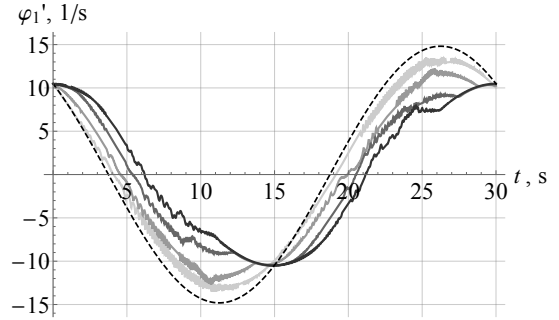


Fig. 16. Dependencies $\dot{\varphi}_1(t)$ case 2 (**model N** — black dashed line; **model CF** — light gray solid line; **model MF**: $\chi = 2$ mm — gray solid line, $\chi = 5$ mm — dark gray solid line, $\chi = 8$ mm — black solid line)

different values of contact spot radius are shown in Fig. 17 (traversing counterclockwise) and Fig. 18.

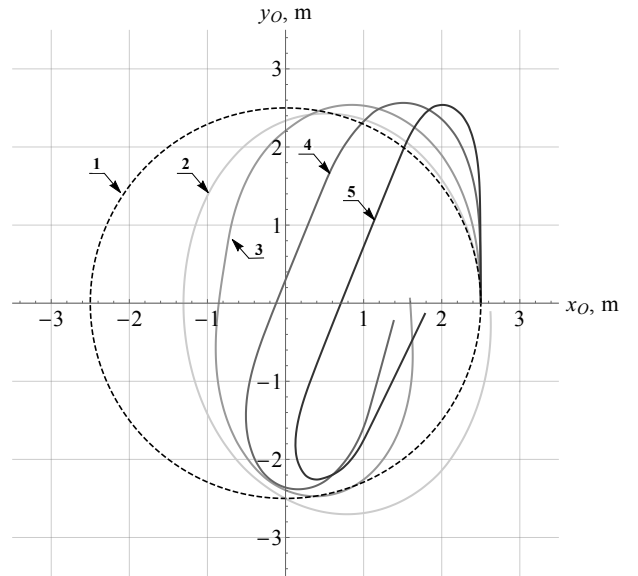


Fig. 17. Trajectories of the center of the platform O case 2 (1 — **model N**; 2 — **model CF**; 3 — **model MF**, $\chi = 2$ mm; 4 — **model MF**, $\chi = 5$ mm; 5 — **model MF**, $\chi = 8$ mm)

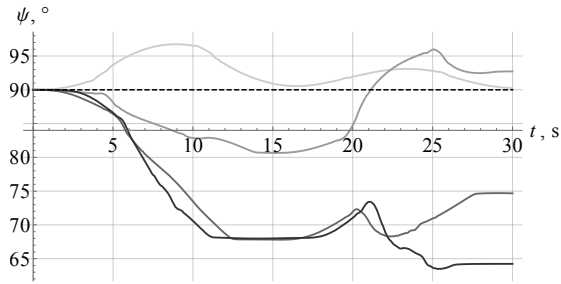


Fig. 18. Rotation angle of the platform ψ **case 2** (**model N** — black dashed line; **model CF** — light gray solid line; **model MF**: $\chi = 2$ mm — gray solid line, $\chi = 5$ mm — dark gray solid line, $\chi = 8$ mm — black solid line)

Using one wheel as an example, we present the dependencies of sliding friction (Fig. 19) and spin friction torque (Fig. 20). Note that the spin friction torque increase with an increase in the radius of the contact spot.

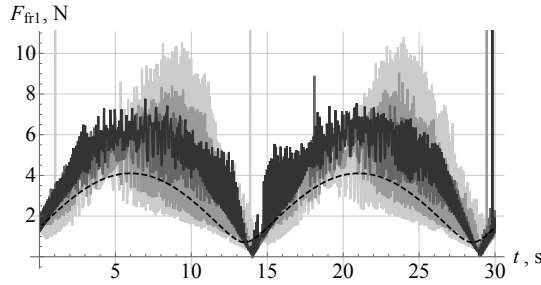


Fig. 19. Dependencies sliding friction force modulus $F_{fr1} = \sqrt{F_{X1}^2 + F_{Y1}^2}$ **case 2** (**model N** — black dashed line; **model CF** — light gray solid line; **model MF**: $\chi = 2$ mm — gray solid line, $\chi = 5$ mm — dark gray solid line, $\chi = 8$ mm — black solid line)

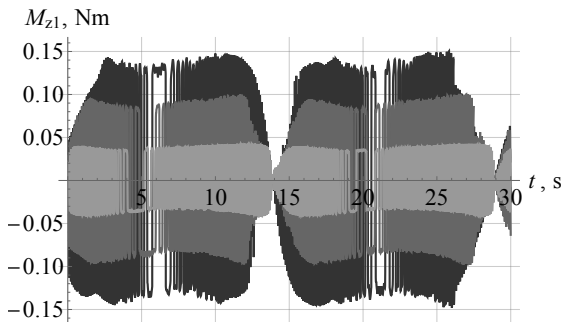


Fig. 20. Dependencies spin friction torque M_{z1} **case 2** according to the simulation results for **model MF** ($\chi = 2$ mm — gray solid line, $\chi = 5$ mm — dark gray solid line, $\chi = 8$ mm — black solid line)

V. CONCLUSION

This study presents dynamic model of the movement of an omni-directional platform, taking into account the design of the wheels, and comparative analysis of simulation results for different values of the parameters of dissipative forces. Two models of contact friction forces are considered: two-dimensional Coulomb friction and the multicomponent friction model that takes into account sliding and spinning.

The results of numerical simulation show that increasing the radius of the contact spot leads to a decrease in the efficiency of controlling the lateral velocity of platform center and, as a result, to an increase in energy consumption for the exact implementation of the reference motion and deformation of the trajectories of the platform points. It is also shown that a decrease in the value of the linear friction parameters leads to a decrease in the values of the control moments predicted by the nonholonomic model for the implementation of the programmed motion, and, as a result, the trajectory of motion is differs significantly from the programmed motion.

The possible significant influence values of parameters of the dissipative forces on the platform movement is shown.

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